# MLDS Project Proposal

## Label-Free Aggregation of Human Annotators with FUSE, and Training on the Result

**Date:** November 2026  **Mentor:** Dr. Regev Cohen

## Background

Supervised models are only as good as their labels, and labels from a single expert are slow, costly, and still imperfect. A common alternative is to collect several cheaper annotations per item and combine them. The standard default is naive majority vote, which treats all annotators as equally reliable; it is the baseline this project aims to beat. Dawid–Skene [3] learns each annotator's reliability by EM but can be unstable when each item has few annotations. FUSE (Fully Unsupervised Score Ensembling) [1] takes a different route: it treats each annotator as a noisy binary classifier, estimates its sensitivity and specificity by the method of moments of Jaffe et al. [2] from the agreement structure among annotators alone, and outputs a posterior probability per item, with no labels at all. FUSE was designed and evaluated for ranking language-model answers using LLM judges as the "annotators." Whether it helps when the annotators are people, and whether the labels it produces train a better model than one expert's labels, is open. This project asks both.

**Preliminaries.** Item $i$ has an unobserved label $y_i \in \{1, \dots, K\}$; annotator $j$ labels a subset of items, giving $A_{ij}$ or a missing entry. In the binary case, annotator $j$ has sensitivity $\psi_j$ and specificity $\eta_j$; under conditional independence given $y$ the off-diagonal covariance of $A$ is rank one with entries $\lambda_j \lambda_k$, $\lambda_j \propto \psi_j + \eta_j - 1$, so each $\lambda_j$ is identifiable from any three annotators. Multi-class tasks are handled one-vs-rest. FUSE's posterior $p_i$ is used as a hard label (arg max) or as a soft training target. A "single expert" means one annotator's label on every training item. Human annotators may violate conditional independence, since they share guidelines, biases and class priors, so independence is a risk to be measured, not assumed.

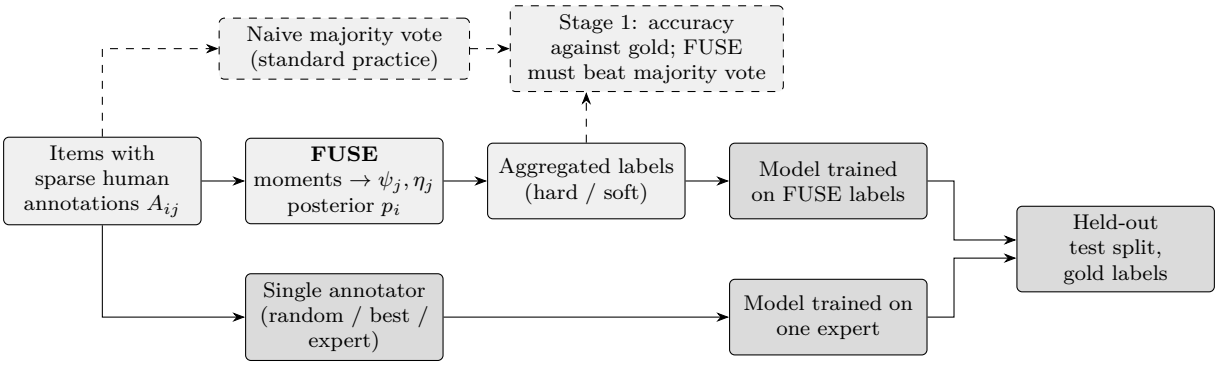



Figure 1: Study design. Light grey boxes belong to Stage 1, dark grey to Stage 2. Stage 1 asks whether FUSE's aggregated labels are closer to gold than those of standard aggregators; Stage 2 compares a model trained on those labels with one trained on a single annotator's labels, on a test split no aggregator touched.

**Stage 1 (until the midterm presentation): does FUSE improve the aggregation decision on human labels?**

a) **Data.** Four to six public datasets with several human annotations per item and a curated gold label: binary (textual entailment, Snow et al. [4]; relevance and sentiment sets from the Zheng et al. benchmark [5]) and multi-class (CIFAR-10H [6]; LabelMe [7]). Gold labels are hidden from every method and used only for scoring.

b) **Implementation.** Implement FUSE from the paper and extend it to what human data needs: missing entries (pairwise-complete moments with a minimum-overlap rule), multi-class, and calibrated posteriors. Unit-test on simulated annotators with known $\psi_j, \eta_j$.

c) **Aggregation benchmark** (mandatory). FUSE against naive majority vote (the standard practice and the main baseline to beat), Dawid–Skene and Jaffe et al., on accuracy and log loss against gold. Annotations per item $k$ are subsampled to trace accuracy as a function of annotation budget.

d) **Stress tests** (secondary). Simulated and subsampled annotator groups with shared bias, spammers and adversarial minorities, to find where the method degrades, and whether a label-free check can flag it.

**Midterm criterion, fixed in advance.** On a dataset, Stage 1 succeeds if FUSE's accuracy gain over majority

vote has a paired-bootstrap 95% interval above zero at a stated $k$; on the study, if this holds on most datasets. "FUSE beats majority vote only in regime X" is a valid outcome, and Stage 2 then uses the best aggregator found.

**Stage 2: does a model trained on FUSE labels beat one trained on a single expert?** A classifier (small CNN on image sets; pretrained encoder on a text set) is trained with identical architecture, optimiser and budget on labels from six arms:

(1) a single annotator, in three versions: a random annotator; the best annotator chosen by gold agreement (an optimistic bound for "one expert"); the dataset's own expert where one exists;
(2) naive majority vote (the standard practice);
(3) Dawid–Skene;
(4) FUSE hard labels;
(5) FUSE soft labels;
(6) gold labels, as the ceiling.

Each arm is scored on a held-out gold-labelled test split no aggregator touched, by accuracy, log loss and calibration error over at least five seeds. Because several crowd labels cost more than one, the comparison is also made at equal annotation cost over a range of assumed expert-to-crowd price ratios, and the ratio where the advantage disappears is reported. *Extension (gated on a Stage 2 gain):* feed the trained model back into FUSE as one more annotator.

**Evaluation discipline.** *Leakage:* gold enters only in scoring, and hyperparameters are tuned on the single-annotator arm so FUSE is never the favoured arm. *Circularity:* aggregated labels train on one item split and are tested on a disjoint one. *Imperfect gold:* gold labels are noisy, so near-ceiling accuracy is not interpreted. *Multiple comparisons:* results are paired bootstrap intervals over items, with every dataset, arm and $k$ reported, not the best cell. *Observational limits:* datasets differ in annotator recruitment, so differences are described, not causally attributed.

## Objectives & Deliverables

**Objectives.** (1) Adapt FUSE to human annotators. (2) Show, against gold and under the criterion above, whether FUSE improves label aggregation over standard methods and at which annotation budgets; this closes Stage 1. (3) Train models on FUSE labels and compare them with models trained on one expert, at equal items and at equal cost; this is the central contribution.

**Deliverables.** An open-source FUSE library with tests for missing data and multi-class tasks; a reproducible benchmark (aggregation accuracy by $k$; Stage 2 metrics by arm with intervals); a midterm presentation evaluating the Stage 1 criterion; a final report stating, falsifiably, when FUSE labels beat a single expert's, when they do not, and the cost ratio at which the answer flips.

## Requirements & Resources

Advanced statistics, Python and a deep-learning framework; about 1–2 days a week for an academic year. Public datasets only; Stage 1 runs on a laptop, Stage 2 on a single GPU. Academic project; any IP will be addressed when such occurs.

## References

[1] Lee, Ma, Zhao, et al. Fully Unsupervised Score Ensembling (FUSE). arXiv:2604.18547.

[2] Jaffe, A., Nadler, B., Kluger, Y. (2015). Estimating the accuracies of multiple classifiers without labeled data. AISTATS.

[3] Dawid, A. P., Skene, A. M. (1979). Maximum likelihood estimation of observer error-rates using the EM algorithm. J. Roy. Stat. Soc. C 28(1), 20–28.

[4] Snow, R., O'Connor, B., Jurafsky, D., Ng, A. Y. (2008). Cheap and fast—but is it good? Evaluating non-expert annotations for natural language tasks. EMNLP.

[5] Zheng, Y., Li, G., Li, Y., Shan, C., Cheng, R. (2017). Truth inference in crowdsourcing: Is the problem solved? PVLDB 10(5).

[6] Peterson, J. C., Battleday, R. M., Griffiths, T. L., Russakovsky, O. (2019). Human uncertainty makes classification more robust. ICCV.

[7] Rodrigues, F., Pereira, F. C. (2018). Deep learning from crowds. AAAI.